

1.4 Implicit differentiation

Before $y = f(x)$ find $\frac{dy}{dx}$

but what if $g(x, y) = 0$ how can we

find $\frac{dy}{dx}$? We can find an implicit expression

Example $x^2 + y^2 = 1$ " $y = y(x)$ "

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}1$$

$$2x + 2y \frac{dy}{dx} = 0$$

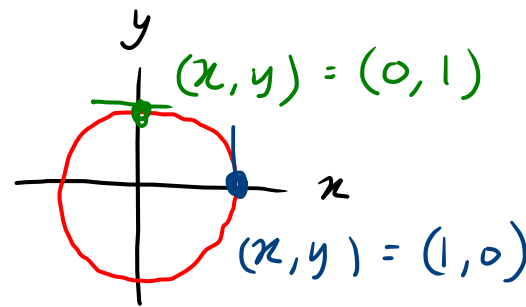
$$\frac{dy}{dx} = -\frac{2x}{2y}$$

Calculate $\frac{dy}{dx}$ at $(x, y) = (0, 1)$

$$\left. \frac{dy}{dx} \right|_{(0,1)} = -\frac{0}{1} = \underline{0}$$

Calculate $\frac{dy}{dx}$ at $(x, y) = (1, 0)$

$$\left. \frac{dy}{dx} \right|_{(1,0)} = -\frac{1}{0} \text{ does not exist}$$



$$x^2 + y^2 = 1 \quad "x = x(y)"$$

$$\frac{d}{dy}(x^2 + y^2) = \frac{d}{dy}1$$

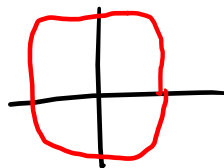
$$2x \frac{dx}{dy} + 2y = 0$$

$$\frac{dx}{dy} = -\frac{y}{x}$$

$$\left. \frac{dx}{dy} \right|_{(1,0)} = -\frac{0}{1} = \underline{0}$$

Second derivatives

Example: $x^4 + y^4 = 1$ ***



$y'' = ?$ Idea: first calculate y' and then y''

$$\frac{d}{dx} x^4 + y^4 = \frac{d}{dx} 1$$

$$4x^3 + 4y^3 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x^3}{y^3}$$

$$y'' = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(-\frac{x^3}{y^3} \right) = - \frac{3x^2 y^3 - x^3 \cdot 3y^2 \frac{dy}{dx}}{y^6}$$

$$= - \frac{(3x^2 y^3 + 3x^6 y^{-1})}{y^6}$$

$$= -3x^2 \frac{(y^4 + x^4)}{y^7} **$$

$$= -\frac{3x^2}{y^7}$$

3.6) Derivatives of logarithmic functions & inverse trigonometric functions

Before $b^x = y$, $(b^x)' = b^x \ln b$

let's try $(\log_b(x))'$ assume that $x, b > 0$

$$\log_b(x) = y \Leftrightarrow b^y = x$$

$$\frac{d}{dx}(b^y) = \frac{dx}{dy} \quad "y = y(x)"$$

$$y' b^y \ln b = 1$$

$$y' = \frac{1}{b^y \ln b} = \frac{1}{x \ln b}$$

$$\underline{(\log_b x)' = \frac{1}{x \ln b}}$$

if $b = e$ what is
 $(\ln x)' = (\log_e x)' = \underline{\frac{1}{x}}$

so how about $x < 0$? Is there an
 $f'(x) = \frac{1}{x}$ for $x \neq 0$

$$(\ln(\underbrace{-x}_{>0}))' = -\frac{1}{-x} = \frac{1}{x} \quad x < 0$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$(\ln|x|)' = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{x} & \text{if } x < 0 \end{cases}$$

$$\ln|x| = \frac{1}{x} \quad \text{for } x \neq 0$$

Logarithmic differentiation!

Example: $\left(n^n \right)' = ?$ if $n > 0$

$$n^n = y \quad *$$
$$\ln(n^n) = \ln(y)$$

$$\frac{d}{dn} n \ln(n) = \frac{d}{dn} \ln(y) \quad " y = y(n) "$$

$$\ln(n) + \frac{n}{n} = y' \frac{1}{y}$$

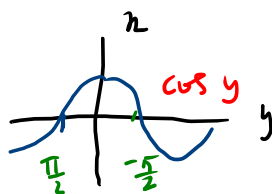
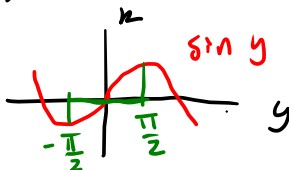
$$\ln(n) + 1 = \frac{y'}{n^n} \quad *$$

$$\underline{n^n (\ln(n) + 1) = y'}$$

Inverse trigonometric functions

Example $y = \sin^{-1}(x) \Leftrightarrow \sin(y) = x$

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



$$(\sin^{-1} x)' = ?$$

$$\sin^{-1} x = y$$

$$\frac{d}{dx} x = \frac{d}{dx} \sin y$$

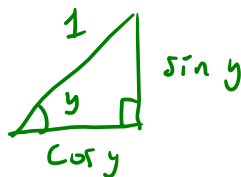
$$y = y(x)$$

$$1 = y' \cos y$$

$$\frac{1}{\cos y} = y'$$

$$y' = \frac{1}{\sqrt{1 - \sin^2 y}}$$

$$\sin(y) = x$$
$$= \frac{1}{\sqrt{1 - x^2}}$$



$$\cos^2 y + \sin^2 y = 1$$

$$\cos^2 y = 1 - \sin^2 y$$

$$\cos y = \sqrt{1 - \sin^2 y}$$

or

$$\cos y = -\sqrt{1 - \sin^2 y}$$

Derivatives of Inverse Trigonometric Functions

$$\rightarrow \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} (\csc^{-1} x) = -\frac{1}{x\sqrt{x^2 - 1}}$$

$$\frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1 - x^2}}$$

$$\frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2 - 1}}$$

$$\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1 + x^2}$$

$$\frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1 + x^2}$$